Brazilian Journal of Psychiatry
bjp
Revista Brasileira de Psiquiatria
Brazilian Psychiatric Association
Fully open access
No APCs

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(article published as accepted)

Brief Communication

Social cognition across bipolar disorder and behavioral variant frontotemporal dementia: an exploratory study

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http://doi.org/10.47626/1516-4446-2022-2935

Submitted: 27-Oct-2022

Accepted: 03-Jan-2023

This is a preliminary, unedited version of a manuscript that has been accepted for publication in the Brazilian Journal of Psychiatry. As a service to our readers, we are providing this early version of the manuscript. The manuscript will still undergo copyediting, typesetting, and review of the resulting proof before it is published in final form. The final version may present slight differences in relation to the present version.

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Social cognition across bipolar disorder and behavioral variant frontotemporal dementia: an exploratory study

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Word count for the title: 13

Characters count for the title (including spaces): 109

Word count for the paper (excluding references and caption): 1563

Word count for the abstract: 170

Number of references: 16

Number of tables: 01

Number of figures: 01

Abstract

Objectives: Bipolar Disorder type 1 (BD1) and behavioral variant frontotemporal dementia (bvFTD) share similar behavioral and cognitive symptoms, rendering the differential diagnosis between them a clinical challenge. We investigated the accuracy of social cognition measures to differentiate bvFTD from BD.

Methods: We included three groups of participants: early-onset BD1 (in remission, n = 20), bvFTD (n = 18), and cognitively healthy controls (HC, n = 40), matched for age, schooling, and sex. All participants underwent cognitive assessment including the facial emotion recognition (FER) and the modified *faux-pas* tests, which assess mentalizing.

Results: Compared to HC, BD1, and bvFTD patients underperformed on both social cognition measures. BD1 and bvFTD did not differ on the FER's or faux pas tests' total score, although bvFTD patients had significantly higher difficulties than BD1 to detect social *faux-pas* ($p < 0.001$, $d = 1.35$).

Conclusions: BD1 and bvFTD share deficits in the core social cognition functions. These findings should be considered in the development of tasks aiming to improve the clinical differentiation between both disorders.

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Introduction

Bipolar Disorder (BD) is a chronic, disabling, and complex mental illness, associated with poor functionality and reduced interpersonal functioning[1]. BD mood swings Patients with BD might exhibit psychiatric and behavioral symptoms that are also criteria for the diagnosis of behavioral variant frontotemporal dementia (bvFTD)[2], including impulsivity, disinhibition and apathy. Besides neuropsychiatric manifestations, BD and bvFTD also share similar cognitive deficits such as executive dysfunction, commonly present in both disorders[3,4]. Social cognition (SC) refers to a set of cognitive functions involved in the processing of social signals and are critical for social adaptation and appropriate interpersonal exchanges. Emotion recognition and mentalizing have been described as its core processes. Impairments of SC are frequent in bvFTD[4], but could be present in BD as well[1, 3].

The differential diagnosis between bvFTD and primary psychiatric disorders (PPD) is a key challenge, particularly in behavioral disorders initiated after the age of 50 years[5]. For most clinicians, the distinction between FTD and BD may be challenging, and a thorough revision of psychopathology is required. BD seems to present more frequently grandiosity, often accompanied by increased energy and decreased need for sleep, disinhibition and euphoric features, and a past history of depression. Instead, some FTD individuals may show and stereotyped and inadequate sexual behavior, such as public masturbation, and may show repetitive motor behavior and stereotypies. bvFTD is the most common cause of young-onset dementia after Alzheimer's disease and frequently manifests with marked behavioral and

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personality changes, such as disinhibition, loss of empathy, and social misconduct. Patients with bvFTD may also present psychiatric symptoms frequently retrieved in PPD such as mania, delusions, and hallucinations[2]. These shared features may led to an important delay in the diagnosis of bvFTD as at least 50% of patients receive a psychiatric diagnosis prior to the correct diagnosis[6]. Almost 25% of patients with bvFTD might present the BD diagnosis during the lifetime[7]. In this context, BD might present as the most challenging differential diagnosis to bvFTD[6].

In the absence of biomarkers, the use of SC tasks might be a tool to help to differentiate these conditions[2]. However, data are critically lacking to support these recommendations, as the studies that have performed a direct contrast between early-onset dementia and late-onset psychiatric conditions are scarce[5]. To date, only one study[5] directly compared SC in patients with BD and bvFTD, and demonstrated poorer performance on a mentalizing test in bvFTD. Here, we aimed to compare the performance in SC' core processes of remitted BD type 1 (BD 1) and bvFTD patients.

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Methods

We included 20 remitted early-onset BD 1 patients, 18 bvFTD patients, and 40 cognitively healthy controls (HC) in the present study. All patients were recruited at the University Hospital of the Federal University of Minas Gerais (Belo Horizonte, Brazil). BD 1 were assessed with the Mini-International Neuropsychiatric Interview; the 17-items Hamilton Depression Rating Scale (HDRS), and the Young Mania Rating Scale (YMRS) to determine the severity of depressive and

manic symptoms. Remission was defined by YMRS and HDRS scores <7 points, for at least 8 weeks.

bvFTD patients fulfilled consensual diagnostic criteria for probable bvFTD [8], *i.e.*, patients had a consistent history of personality and behavioural changes associated to functional decline and evidence of focal atrophy or hypoperfusion in frontotemporal regions on brain magnetic resonance imaging (MRI) or on single-photon emission computed tomography (SPECT), respectively. The previous diagnosis of BD or schizophrenia was an exclusion criterion for this group. HC were recruited from the local population and had no history of psychiatric or neurologic disorders. They did not present cognitive complaints and had normal scores on the Mini-Mental State Examination (MMSE).

All participants underwent a cognitive evaluation that included the MMSE, the Frontal Assessment Battery (FAB), and the Figure Memory Test (FMT) from the Brief Cognitive Screening Battery (BCSB).

SC was assessed with the mini Social cognition and Emotional Assessment (Mini-SEA), composed of a Facial Emotion Recognition (FER) and the modified *faux-pas* (mFP) tests assessing mentalizing. For the mFP, in addition to the total score, subscores were also considered: detection, attribution, identification, knowledge, intentionality, and empathy. Moreover, two components of the test were distinguished: the correct recognition of social norms violation (the sum of scores for questions 1, 2 and 3); and mentalizing ability (the sum of scores for questions 4, 5 and 6). The total Mini-SEA score is the sum of the FER and mFP scores, with higher scores indicating better performance.

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The local Ethics Research Committee approved the study. All participants or their legal representatives provided written informed consent.

Statistical Analyses

Statistical analyses were conducted using SPSS 22.0 (SPSS Inc., Chicago, Illinois). Shapiro-Wilk test refuted the assumption of normality. Non-parametric Kruskal-Wallis, Mann-Whitney tests & effect size (Cohen's *d*) were used for group comparisons. We adopted Bonferroni's correction for multiple comparisons; the level of significance (α) was set at 0.025. The accuracy of SC measures was investigated with Receiver Operating Characteristics (ROC) curve analyses.

Results

Table 1 presents the demographic, clinical and neuropsychological data. Participants did not differ on age, sex and schooling. Compared to HC, bvFTD and BD1 patients presented poorer performances on all cognitive measures. Remitted BD 1 and bvFTD patients had similar performance on all SC scores, except on the detection of *faux-pas* (to identify whether or not there is a *faux-pas)*, with bvFTD performing worse than BD1 ($p<0.001$, $d=1.29$) (Figure 1). Results were similar when controlling for executive performance (FAB).

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Table 1 Demographical and neuropsychological data (mean ± standard deviation) for the study population

	bvFTD (n = 18)	**BD (n = 20)**	**Healthy Controls (n = 40)**	**Kruskal-Wallis Test (all groups)**	**bvFTD vs BD (Mann-Whitney test)**	**bvFTD vs controls (Mann-Whitney test)**	**BD vs controls (Mann-Whitney test)**
Sex (Female/Male)	9F/9M	12F/8M	27F/13M	Not applicable	p = 0.51 ¥	p = 0.07 ¥	p = 0.02 ¥
Disease duration (years)	3.6 (± 2.1)	32.1 (±13.1)	NA	Not applicable	p < 0.0001, *d* = 3.04	Not applicable	Not applicable
Age (years)	65 ± 8.8	57.4 ± 11.4	61.2 ± 9.9	p = 0.12	p = 0.04 ; *d* = 0.76	p = 0.21, *d* = 0.42	p = 0.23 *d* = 0.35
Educational level (years)	12.7 ± 3.2	12.0 ± 2.7	12.5 ± 2.9	p = 0.91	p = 0.88, *d* = 0.21	p = 0.93, *d* = 0.03	p = 0.63, *d* = 0.02
Mini-Mental State Exam (/30)	26.1 ± 1.9	27.1 ± 1.7	28.9 ± 1.2	p < 0.001	p = 0.65, *d* = 0.55	p = 0.38, d = 1.84	p = 0.0001; *d* = 1.26

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Frontal Assessment Battery (/ 18)	11.7 ± 3.6	13.7 ± 3.7	16.3 ± 1.6	p < 0.001	p = 0.07, *d* = 2.65	p < 0.0001; *d* = 1.59	p < 0.0001 , *d* = 1.15
Figure Memory Test – 5 minutes Recall	6.7 ± 2.4	7.57 ± 2.3	9.27 ± 1.4	p < 0.001	p = 0.35, *d* = 0.47	p < 0.0001; *d* = 1.31	p < 0.002; *d* = 0.92
Facial Emotion recognition test (/35)	20.6 ± 6.3	22.6 ± 4.4	28.7 ± 2.4	p < 0.001	p = 0.03, *d* = 0.59	p = 0.01 ; *d* = 0.76	p < 0.0001 ; *d* = 1.73
Theory of mind: Faux-pas Recognition Test							
Total score (/40)	20.4 ± 7.4	25.7 ± 8.4	35.9 ± 3.8	p < 0.001	p = 0.20; *d* = 0.53	p < 0.0001; *d* = 2.65	p < 0.0001 , *d* = 1.55
Detection of faux-pas (/5)	3.3 ± 1.3	4.7 ± 0.6	4.9 ± 0.4	p < 0.001	p < 0.001; *d* = 1.29	p < 0.0001; *d* = 1.54	p = 0.23 ; *d* = 0.39
Detection of absence of faux-pas (/5)	4 ± 1.0	3.4 ± 1.4	4.8 ± 0.6	p < 0.001	p = 0.27 ; *d* = 0.50	p < 0.0001; *d* = 0.95	p < 0.0001 ; *d* = 1.27
Attribution (/10)	2.9 ± 1.4	3.3 ± 1.4	4.8 ± 1	p < 0.001	p = 0.44 ; *d* = 0.27	p = 0.38 ; *d* = 1.59	p < 0.001 ; *d* = 1.25

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Identification (/10)	2.1 ± 2.0	2.4 ± 1.6	4.3 ± 0.9	$p < 0.001$	$p = 0.42$, $d = 0.21$	$p < 0.0001$; $d = 1.51$	$p < 0.0001$; $d = 1.41$
Knowledge (/10)	0.9 ± 1.0	1.8 ± 1.5	3.8 ± 1	$p < 0.001$	$p < 0.12$, $d = 0.55$	$p < 0.0001$, $d = 2.37$	$p < 0.0001$, $d = 1.5$
Intentionality (/10)	2.6 ± 1.4	2.4 ± 1.5	4.0 ± 0.9	$p < 0.001$	$p < 0.42$, $d = 0.27$	$p < 0.001$, $d = 1.04$	$p < 0.0001$, $d = 1.29$
Empathy (/10)	2.6 ± 1.2	3.3 ± 1.5	4.6 ± 0.7	$p < 0.001$	$p < 0.25$, $d = 0.44$	$p < 0.0001$, $d = 1.82$	$p < 0.001$; $d = 1.13$
Recognition of social norms violation (/20)	12 ± 4.6	13.8 ± 4.3	18.7 ± 1.9	$p < 0.001$	$p < 0.27$, $d = 0.38$	$p < 0.38$, $d = 1.89$	$p < 0.0001$; $d = 1.48$
Mentalizing (/20)	6.5 ± 3.5	7.4 ± 4.2	12.4 ± 2.1	$p < 0.001$	$p < 0.65$, $d = 0.23$	$p < 0.38$, $d = 2.02$	$p < 0.0001$; $d = 1.47$
Mini-SEA Total Score (/30)	17.1 ± 4.8	19.3 ± 4.3	25.7 ± 1.9	$p < 0.001$	$p < 0.28$, $d = 0.50$	$p < 0.0001$, $d = 2.44$	$p < 0.0001$; $d = 1.92$

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Legend: BD: remitted Type I Bipolar Disorder; bvFTD: Behavioral Variant Frontotemporal Dementia; *d*: Cohen's *d*; Mini-SEA: Short version of Social and Emotional Assessment; NA: Not applicable

¥ Chi-square test

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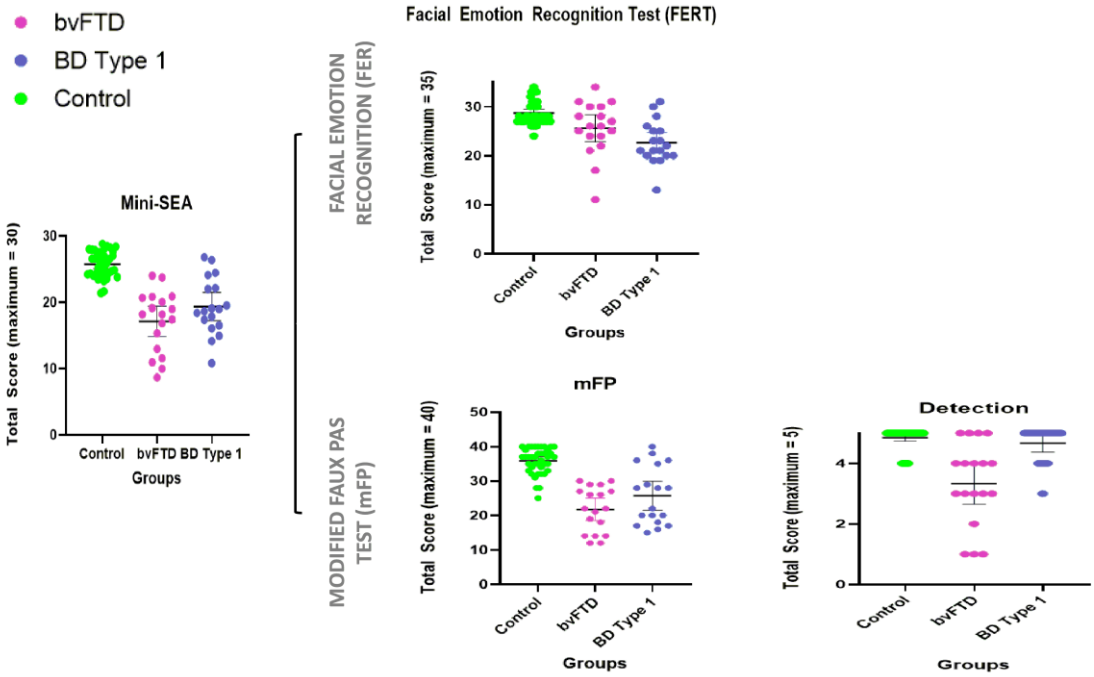

Figure 1 Social Cognition tests. Individual and group performance of controls and patients with behavioural variant frontotemporal dementia (bvFTD) and remitted Type I Bipolar Disorder (BD) at the short version of Social and Emotional Assessment (Mini-SEA), which is composed of a facial emotion recognition test (FERT) and the modified faux pas (mFP) test.

The area under the curve (AUC) values (BD vs bvFTD) for mFP's, FER's and Mini-SEA's total scores were 0.61 (sensitivity=0.78, specificity=0.28), 0.29 (sensitivity =0.67, specificity=0.12), and 0.63 (sensitivity =0.78, specificity =0.34), respectively. For the detection of *faux-pas*, the AUC value was 0.80 (sensitivity=0.72, specificity=0.78).

Discussion

This is the first study to compare the performance on SC' core processes (facial emotion recognition and mentalizing) between remitted BD1 and bvFTD patients. These clinical groups did not differ in all cognitive and SC measures, except for the detection of *faux-pas*.

BD1 and bvFTD had lower scores than HC on MMSE, FAB and episodic memory scores, but did not differ between each other. Executive and episodic memory impairment are commonly observed in bvFTD[4], and might be present in BD[3]. Our findings highlight that standard cognitive measures do not contribute to the diagnostic differentiation between bvFTD and BD.

Clinical follow-up is typically envisaged as useful to distinguish both conditions in the challenging context of global cognitive deterioration as longitudinal studies demonstrated cognitive performance remained stable in BD over a 6-year period[9,10]. Recent systematic review suggested that BD is associated with a three-fold increased risk of dementia, although authors appointed high heterogeneity of studies might be an important factor with the low strength of evidence[11]. Interestingly, some case reports suggested an association between bvFTD and BD[12]. This observation highlights the need to find specific cognitive markers to reliably distinguish both disorders[2].

As expected, BD and bvFTD had lower performance than HC in emotion recognition and mentalizing, two core dimensions of SC, in agreement with previous studies[1,3,4]. A unique study[5] reported that bvFTD patients had lower scores than BD (types I and II) on the Reading the Mind in the Eyes, which is controversial regarding the necessary involvement of mentalizing abilities (*vs.* emotion recognition) in the task's performance. In our study, aiming to assess the core components of SC, we employed the mFP, which does involve both mentalizing and social

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context decoding abilities, alongside the FER. Our methodology also relied on strict correction (Bonferroni) for multiple comparisons, thus increasing the reliability of our findings. Our results suggest that the decreased ability to detect *faux pas* could be considered as a specific marker of bvFTD – which is in line with previous studies[13,14]. Interestingly, while BD patients are able to detect *faux pas*, they were unable to explain the *faux pas* correctly, similarly to bvFTD patients. Future studies should explore the qualitative differences in the responses given as potential indicators of diagnosis.

BD and bvFTD patients did not differ on the FER. Previous data show that bvFTD[4] and BD[15] have impairment on emotion recognition although, to the best of our knowledge, this is the first study to directly compare BD and bvFTD on this ability. Previous study demonstrated that FER might differentiate bvFTD and depressive disorders[16], but more studies are warranted to investigate the utility of such tests to differentiate bvFTD from PPD.

We acknowledge that sample sizes as well as narrowing the assessment of executive functions to the FAB are limits to this study. It is therefore not excluded that a more accurate distinction between BD and bvFTD could be provided by an extended cognitive and SC assessment. The effect of medications on cognitive performance was not considered here, which might also be an issue given the use of antipsychotics and mood stabilizers drugs in both disorders, although this is a common limit in the field.

In the absence of biomarkers, the distinction between bvFTD and BD remains a challenge. Altogether, our results should be considered in the development of tasks aiming to improve the clinical differentiation between bvFTD and BD. While the use of SC tests has been recently consensually recommended in this context[2], our study emphasized the importance of

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multidimensionality in this assessment. Beyond the need to assess several functions rather than a single one, our findings also underline the utility of a more qualitative approach in the assessment of SC as differences were retrieved in a single dimension of the mFP, but not the total score.

Acknowledgments

We are grateful to the patients and their families for their contribution to this study.

This study was partially funded by Conselho Nacional de Pesquisa Científica (CNPq), Brazil: Grant 402853/2012.

ALT, LCS and PC are supported by the Brazilian National Council for Scientific and Technological Development (CNPq – *Bolsa de Produtividade em Pesquisa*).

Authors' contributions

IGB designed the study, performed statistical analyses, collected clinical and cognitive data, and critically reviewed the manuscript for intellectual content.

FMCL collected clinical and cognitive data and critically reviewed the manuscript for intellectual content.

MB performed statistical analyses and critically reviewed the manuscript for intellectual content.

LBG, LIM, HCG, and PC collected cognitive/clinical data and critically reviewed the manuscript for intellectual content.

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ALT critically reviewed the manuscript for intellectual content.

LCS designed the study, performed statistical analyses, collected clinical and cognitive data, and drafted the first version of the manuscript.

All authors read and approved the final manuscript.

Disclosures

The authors declare no conflicts of interest.

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